

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

TRAINING TR-102 REPORT DAY 13

9 July 2025

Overview:

In today's session, we learned about Linear Regression, one of the most basic and widely used algorithms in supervised machine learning. It helps to predict outcomes based on input data using a straight-line relationship. This topic marked our first step into applying mathematics and statistics to real-world data prediction.

Learning Objectives:

- Understand the concept of linear regression and its importance
- Learn how a line of best fit is drawn through data
- Explore the role of slope (m) and intercept (b) in prediction
- Know how loss functions and gradient descent improve predictions

Linear Regression

Linear regression is a type of supervised machine-learning algorithm that learns from the labelled datasets and maps the data points with most optimized linear functions which can be used for prediction on new datasets. It is used to find the linear relationship between a dependent variable (target) and one or more independent variables (features). The goal is to fit the best straight line through the data points which can be used for prediction.

For example, we want to predict a student's exam score based on how many hours they studied. We observe that as students study more hours, their scores go up. In the example of predicting exam scores based on hours studied. Here are two types of variables:

- **Independent variable (input):** Hours studied because it's the factor we control or observe.
- **Dependent variable (output):** Exam score because it depends on how many hours were studied.

Equation of Linear Regression:

The line is expressed as:
 $y = mx + b$

where:

- y is the predicted value
- x is the input feature
- m is the slope
- b is the intercept

Best Fit Line in Linear Regression

The goal of linear regression is to find a straight line that minimizes the error between the observed data points and the predicted values. This line helps us predict the dependent variable for new and unseen data.

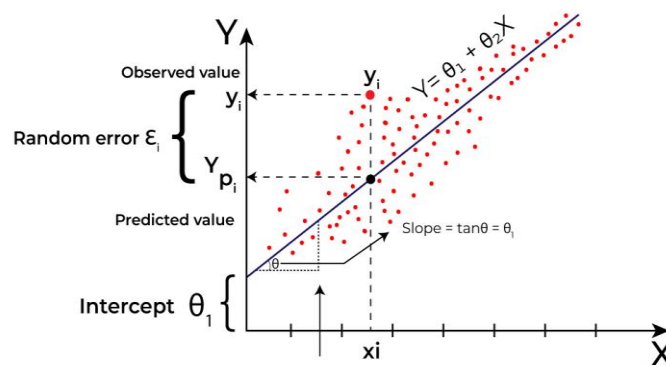


Fig. Linear Regression

Y is called a dependent or target variable and X is called an independent variable also known as the predictor of Y. There are many types of functions or modules that can be used for regression. A linear function is the simplest type of function. X may be a single feature or multiple features representing the problem.

Loss Function & Gradient Descent:

We also learned how a loss function measures the error between the predicted value and actual value. Then, gradient descent is used to update m and b to reduce that error, helping the line better fit the data over time.

Conclusion:

Linear regression gave us a clear understanding of how machines can make predictions using simple math. It is a beginner-friendly yet powerful concept, and it lays the foundation for more complex algorithms. Working on visualizing the data and seeing how the line fits was both educational and satisfying.